

2Q Call Flow Simulator Mobile Computing

Android

app

Gradle Scripts

activity_main.xml

MainActivity.kt

CallFlowAdapter.kt

```
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2
3 import androidx.recyclerview.widget.RecyclerView
4
5 class CallFlowAdapter(private val steps: List<CallFlowStep>) : RecyclerView.Adapter<CallFlowAdapter.ViewHolder>() {
6
7     override fun onCreateViewHolder(parent: ViewGroup, viewType: Int): ViewHolder {
8         val view = LayoutInflater.from(parent.context).inflate(R.layout.item_flow_step, parent, false)
9         return ViewHolder(view)
10    }
11
12    override fun onBindViewHolder(holder: ViewHolder, position: Int) {
13        holder.bind(position)
14    }
15
16    override fun getItemCount(): Int {
17        return steps.size
18    }
19
20    class ViewHolder(view: View) : RecyclerView.ViewHolder(view) {
21        private val tvTitle: TextView = view.findViewById(R.id.tv_title)
22        private val tvDescription: TextView = view.findViewById(R.id.tv_description)
23        private val nodes: List<TextView> = listOf(
24            view.findViewById(R.id.node0),
25            view.findViewById(R.id.node1),
26            view.findViewById(R.id.node2),
27            view.findViewById(R.id.node3),
28            view.findViewById(R.id.node4),
29            view.findViewById(R.id.node5)
30        )
31        private val links: List<View> = listOf(
32            view.findViewById(R.id.link0),
33            view.findViewById(R.id.link1),
34            view.findViewById(R.id.link2),
35            view.findViewById(R.id.link3),
36            view.findViewById(R.id.link4)
37        )
38    }
39
40    fun ViewHolder.bind(position: Int) {
41        tvTitle.text = steps[position].title
42        tvDescription.text = steps[position].description
43        for (i in 0 until nodes.size) {
44            nodes[i].text = steps[position].nodes[i]
45        }
46        for (i in 0 until links.size) {
47            links[i].text = steps[position].links[i]
48        }
49    }
50}
```

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Step 1: Call Initiation

Mobile A initiates a call by sending a request to the Base Station.

Mobile A

Base Station

Radio Access Network

Core Network

Destination Network

Mobile B

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9         return ViewHolder(view)
10    }
11
12    override fun onBindViewHolder(holder: ViewHolder, position: Int) {
13        val step = steps[position]
14        holder.tvTitle.text = step.title
15        holder.tvDescription.text = step.description
16
17        // Bind nodes
18        val nodes = listOf<TextView>{
19            view.findViewById(R.id.node0),
20            view.findViewById(R.id.node1),
21            view.findViewById(R.id.node2),
22            view.findViewById(R.id.node3),
23            view.findViewById(R.id.node4),
24            view.findViewById(R.id.node5)
25        }
26
27        // Bind links
28        val links = listOf<View>{
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30            view.findViewById(R.id.link1),
31            view.findViewById(R.id.link2),
32            view.findViewById(R.id.link3),
33            view.findViewById(R.id.link4)
34        }
35    }
36}
```

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Step 2: Network Access

The Base Station forwards the request through the Radio Access Network to the Core Network.

Mobile A

Base Station

Radio Access Network

Core Network

Destination Network

Mobile B

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7     2 Usages
8
9     class ViewHolder(view: View) : RecyclerView.ViewHolder<View>() {
10
11         4 Usages
12         val tvTitle: TextView = view.findViewById(R.id.tvTitle)
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22             view.findViewById(R.id.node5)
23         }
24
25         2 Usages
26         val links = listOf<View>{
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28             view.findViewById(R.id.link1),
29             view.findViewById(R.id.link2),
30             view.findViewById(R.id.link3),
31             view.findViewById(R.id.link4)
32         }
33     }
34 }
```

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Step 3: Authentication

The Core Network authenticates the user and sends a response back to Mobile A.

Mobile A

Base Station

Radio Access Network

Core Network

Destination Network

Mobile B

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33         }
34     }
35 }
```

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Step 4: Call Setup

The Core Network routes the call to the Destination Network, which alerts Mobile B.

Mobile A

Base Station

Radio Access Network

Core Network

Destination Network

Mobile B

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35}
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Step 5: Communication

The call is established, and voice/data flows between Mobile A and Mobile B.

Mobile A

Base Station

Radio Access Network

Core Network

Destination Network

Mobile B

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```

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Step 6: Call Termination

One of the parties ends the call, and network resources are released.

Mobile A

Base Station

Radio Access Network

Core Network

Destination Network

Mobile B

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